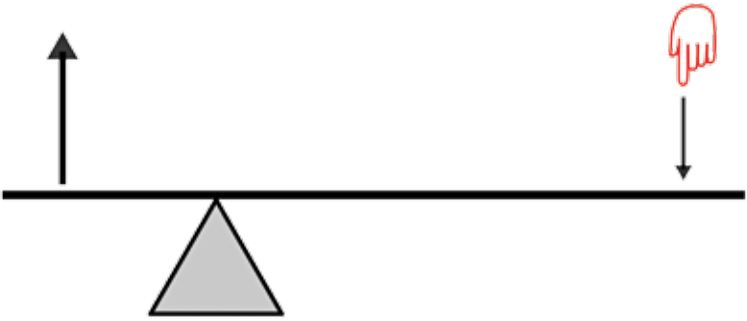


- Processes data and works autonomously to some extent
- Capable of making changes to its physical environment



A lever is one of the simplest machines. These six simple machines form the basis for all mechanical tools, complex machines and robots.

The main difference between a machine and a robot is that robots are ‘smarter’ in the sense that a machine is capable of only performing a set of specific tasks or motions while a robot takes input from its surrounding, processes the data, and then interacts with the environment. A robot can be programmed to perform a variety of tasks and is capable of taking decisions about its actions based on a number of factors.

It might seem surprising to some, but there is still debate about the exact definition of what a robot is and there’s a lot of ambiguity surrounding the word. A toaster is a simple machine because it is capable of performing only a certain task and does not take decisions of its own, while a 3D printer can be called a robot, as it takes input about the design and then processes it to determine how exactly to print the model.

Not every machine can be called a robot while every robot will have some or the other element of these simple machines.

Characteristics of a robot

A machine needs to have some defining characteristics before being counted as a robot. Following are the basic defining characteristics of a robot that a machine must have to be called a robot:

- **Sense** – The machine should be capable of interacting with its environment and sense its surrounding. This can be done using any number of sensors such as light sensors, chemical sensors, pressure and touch